

Scalability & Future Plan

Date	15 March 2024
Team ID	
Project Name	OptiCrop
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Farmer manually enters weather values	Extra effort, possible input errors	High
2	Model limited to 22 crops in dataset	Cannot recommend other crops	Medium
3	Single-region soil assumptions	Lower accuracy in other regions	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Vercel serverless auto-scales	Add caching / CDN for high traffic
2	Data Storage	Static CSV + pickled model	Move to a database + model registry
3	Performance	~150 ms single request	Batch predictions & async endpoints
4	Security	Public, no auth	Add optional accounts & saved fields

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Live weather-API integration (auto-fill)	Next 1-2 months	Removes manual weather entry

Phase 2	Fertilizer recommendation from NPK gaps	3-4 months	Higher yields & savings
Phase 3	Regional soil datasets & retraining	4-6 months	Better nationwide accuracy
Phase 4	Mobile app + multi-language support	6+ months	Wider farmer adoption